

SEPM EXP 1

At its core Devops is the combination of cultural philosophies, practices & tools designed to increase an organization's ability to deliver applications & services at high velocity. By bringing development and operations teams together, organizations can evolve and improve products at a much faster pace than traditional software development models allow.

1. Core Principles

Devops can be best summarized by the CALMS framework.

- **Culture**: The foundation of Devops is collaboration, sharing a common goal and treating product stability and development as everyone's job, not just one team's responsibilities.
- **Automation**: By eliminating repetitive manual work, teams create reliable systems and catch bugs/security flaws earlier in the development lifecycle.
- **Lean**: A DevOps mindset embraces failure and continuously looks for opportunities to improve. Teams assume things will occasionally go wrong so systems are built for fast detection & rapid recovery.
- **Measurement**: Teams track critical data to make decisions such as how long it takes from development to deployment, how often failures occur and how long it takes to recover.
- **Sharing**: Successes and rough patches are shared equally. Pairing Developers & operators throughout the applications lifecycle builds trust and enables better knowledge transfer.

2 DevOps Practices

The technical execution of DevOps relies on a heavily automated pipeline where the order of operations is critical.

i. Continuous Development

The initial phase involves planning the software's functionality, writing the code and utilizing version control systems to manage the codebase.

ii. Continuous Integration

Every time a developer commits a new revision to the codebase, it automatically triggers a build & unit test sequence ensuring bugs are found & addressed much quicker.

iii. Continuous Testing

Automated tests are continuously and repeatedly executed against the codebase across various deployments to ensure quality.

iv. Continuous Delivery & Deployment

Because code is continuously built and tested, good code becomes part of the final product. In continuous delivery the final step is manual whereas in continuous deployment it is automatic.

v. Continuous Monitoring

Monitoring technologies are incorporated to continuously ensure the performance, reliability and health of application and infrastructure in a production environment.

3 Dev Ops Engineer

A devOps Engineer acts as the bridge b/w software creation and software stability. They are tasked with ensuring the pipeline runs efficiently and securely.

Key Responsibilities

- Managing and automating the CI/CD pipelines
- Managing infrastructure and assessing performance
- Scripting, coding and writing specifications & documentation.
- Monitoring System health.

Required Skills & Experience

- **Technical Knowledge:** Proficiency in open-source automation tools (like Git, Jenkins, Docker), cloud services (AWS, Google cloud, Azure) and databases (SQL and noSQL)
- **Coding & Scripting:** Strong grasp of programming languages (Java, C++, Python) and scripting languages (bash, powershell)
- **Background:** Typically requires a BS/MS in computer science, 2+ years of experience as a developer or system administrator and strong data management capabilities.

Beyond technical ~~skill~~ skills, successful devOps engineers need strong communication and leadership abilities, highly developed analytical skills, empathy for other team members and the ability to thrive in cross functional teams.